Annex II. Debt Sustainability Analysis[1]

This Debt Sustainability Analysis (DSA) updates the analysis presented to the Board in July 2016 as background for the 2016 Article IV Report. Liberia's risk of debt distress remains moderate but debt vulnerabilities have risen. The debt-to-exports and debt-to-GDP ratios have deteriorated compared to the 2016 Article IV, reflecting weaker-than-expected exports and growth outlook.

A. Underlying Assumptions

1. The baseline macroeconomic projections in the near- to-medium-term have deteriorated from July 2016 DSA (Text table 1). The economic recovery has not yet materialized after the commodity price shock and the Ebola epidemic. GDP growth has been revised down to an average of 3.8 percent during 2016-2020, from 4.7 percent in the Article IV. Export growth is slower than previously projected. Projected export growth for 2016-2020 is now down from -2.3 percent to -2.4 percent.

Text Table 1. Underlying DSA assumptions

	2016-2017		2016-2020	
	July 2016 DSA	Current DSA	July 2016 DSA	Current DSA
Real GDP growth (percent)	3.8	1.9	4.7	3.8
GDP deflator (annual percent change)	2.0	2.8	3.0	3.5
Primary deficit (percent of GDP)	6.7	6.2	5.5	4.6
Exports growth (percent)	-9.9	-11.0	-2.3	-2.4

Sources: Liberian authorities; and IMF staff projections

B. External DSA

2. The debt distress risk remains moderate under the baseline scenario. The PV of debt-to-GDP ratio in the baseline scenario is projected to increase from 18 percent in FY2016 to 23 percent in FY2020, and gradually decline thereafter. The commodity price shock has exacerbated the exports outlook. As a result, the PV of debt-to-export ratio is projected to rise from 54 percent in FY2016 and to a peak of 99 percent in FY2019, close to the threshold of 100 percent, despite a back-loading of the projected disbursement schedule to reflect the current slower-than-expected pace of disbursements. This peak of 99 percent in the PV of debt-to-export ratio is higher than previously projected (96 percent in July 2016 DSA), mainly due to lower-than-expected export growth. However, the probability charts do not show a breach of thresholds under the baseline scenario and confirm a moderate risk of debt distress.

3. However, debt vulnerabilities remain high, with some breaches under the extreme shock scenario. Under the extreme shock scenario, both the PV of debt-to-GDP ratio and the PV of debt-to-exports ratio breach their thresholds around FY2018–19. These breaches underline the fact that as a commodity exporter, the Liberian economy continues to be sensitive to the external commodity price shock.

[1] Prepared by Atsushi Oshima and Ippei Shibata.

Text Table 2. Composition of External Debt Stock

	End of September 2016	
	Millions of US dollar	Percent of Total
Total debt stock	**597**	**100**
By creditors		
Multilateral including IMF	565	95
Of which:		
World Bank	239	40
AfDB	63	11
Bilateral	33	5

Sources: Liberian authorities; and IMF staff calculations

4. **The authorities agreed with staff's assessment of the moderate risk rating and share staff's concerns about debt vulnerabilities**. They underlined Liberia's challenges of strengthening much-needed infrastructure while preserving debt sustainability. To keep the debt distress risk at moderate, they intend to continue prioritizing potential loans, respect the debt limits under the ECF, which they see as a very useful to guide a prudent borrowing behavior, and limit signature of new loans. The composition of external debt stock comprises mostly multilateral loans (Text Table 2). They also plan to prioritize grants and concessional financing and undertake diversification of the economy to be more resilient to external shocks. Nevertheless, they noted that higher investment would increase growth dividends and in turn help debt sustainability.

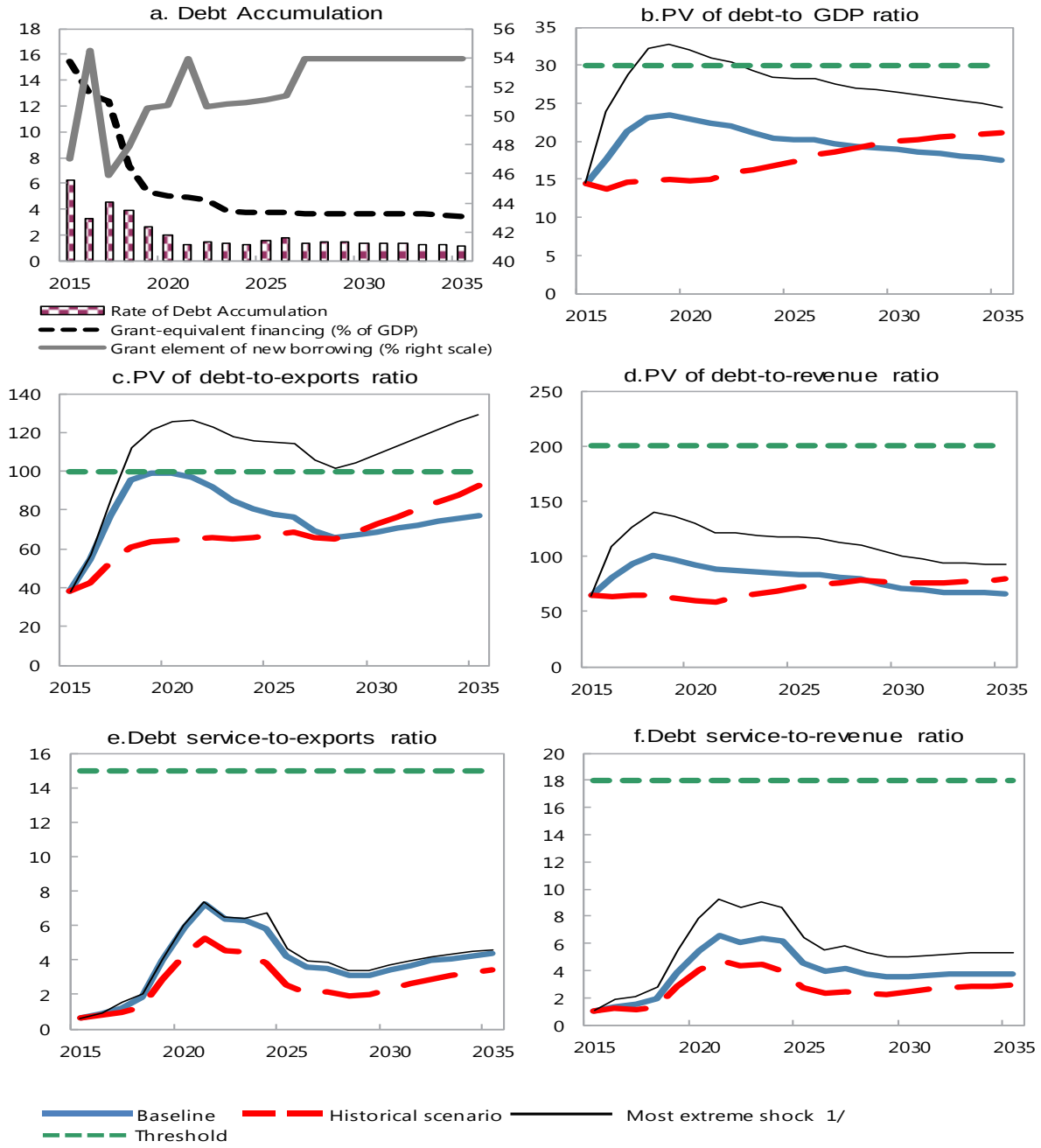
C. Public DSA

5. **The public DSA has not significantly changed compared to the July update**. Under the baseline and shock scenarios, debt indicators remain below the threshold. The PV of public debt-to-GDP ratio is expected to rise from 18 percent in FY2016 to around 29 percent in FY2018 and slowly decline thereafter. The PV of debt-to-revenue ratio is expected to rise from 81 percent in FY2016 to a peak of 100 percent in FY2018 and slowly fall afterwards. The PV of debt service-to-revenue ratio is expected to reach a peak of 6 percent in FY2020 and decline thereafter.

D. Conclusion

6. **Rising debt vulnerabilities call for prudent borrowing policy as well as increased growth and economic diversification**. The DSA shows that Liberia's risk of debt distress is still at a moderate level. However, the commodity price shock and the Ebola epidemic, combined with the UNMIL withdrawal, worsened the growth prospects for the economy and the export industries in particular, deteriorating the debt-to-GDP and debt-to-export ratios compared to the previous DSA. The authorities need to continue pursuing a prudent borrowing strategy while focusing on prioritizing more pro-growth projects, and diversifying the economy to make it more resilient to external shocks.

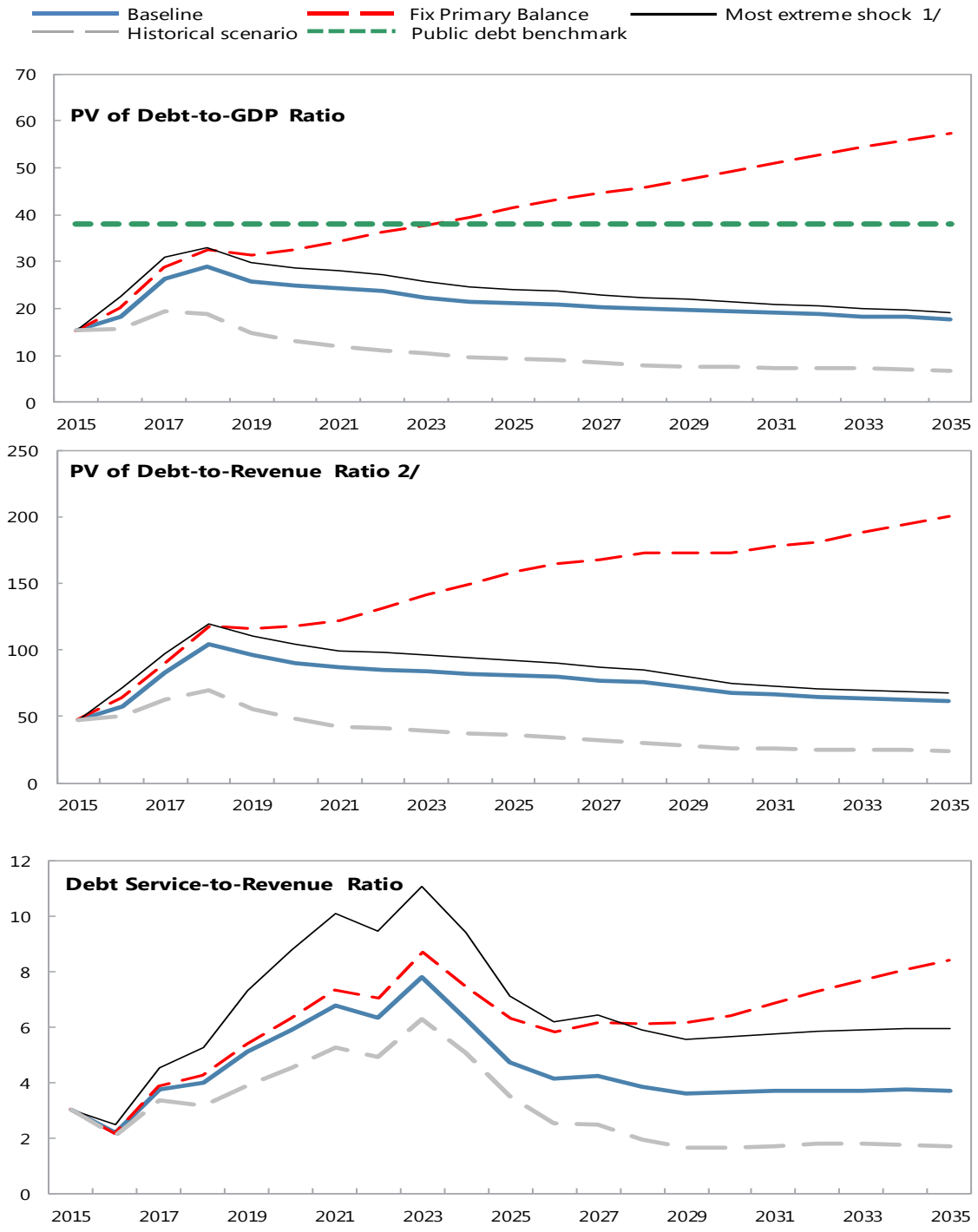
Figure A1. Liberia: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2015–35[1/]



Sources: Country authorities; and staff estimates and projections.

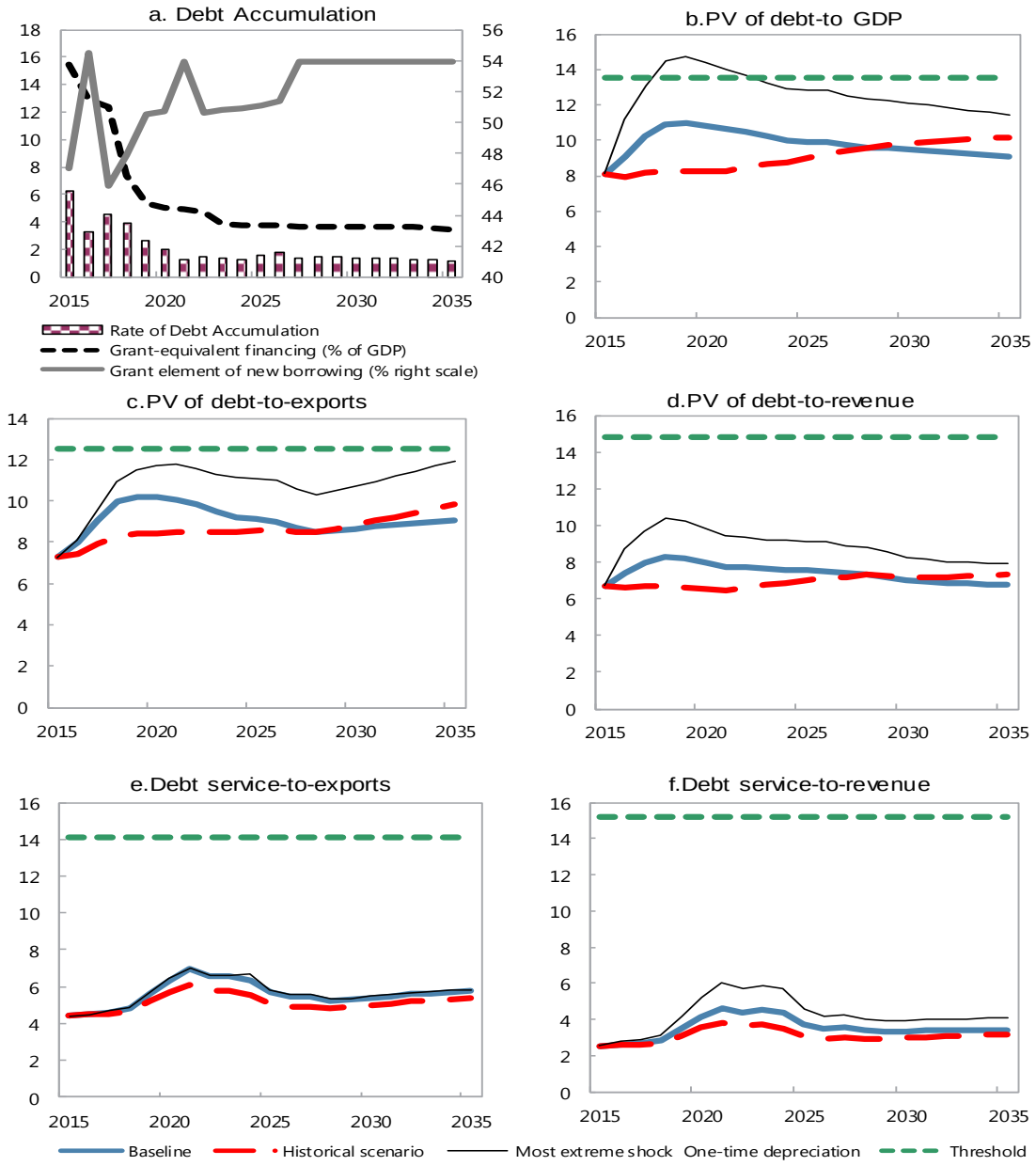
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a One-time depreciation shock; in e. to a Non-debt flows shock and in figure f. to a One-time depreciation shock

Figure A2. Liberia: Indicators of Public Debt under Alternative Scenarios, 2015–35[1/]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2025.
2/ Revenues are defined inclusive of grants.

Figure A3. Liberia: Probability of Debt Distress of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2015–35[1/]



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2025. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a One-time depreciation shock; in e. to a Non-debt flows shock and in figure f. to a One-time depreciation shock

Table A1. Liberia: External Debt Sustainability Framework, Baseline Scenario, 2012–35[1/]

(Percent of GDP, unless otherwise indicated)

	Actual			Historical[6/] Average	Standard[6/] Deviation	Projections									
	2012	2013	2014			2015	2016	2017	2018	2019	2020	2015-2020 Average	2025	2035	2021-2035 Average
External debt (nominal) 1/	**9.4**	**10.9**	**13.6**			**23.4**	**29.4**	**35.1**	**38.0**	**38.5**	**37.9**		**35.0**	**31.1**	
of which: public and publicly guaranteed (PPG)	9.4	10.9	13.6			23.4	29.4	35.1	38.0	38.5	37.9		35.0	31.1	
Change in external debt	-2.2	1.5	2.7			9.8	6.0	5.7	3.0	0.4	-0.5		-0.3	-0.5	
Identified net debt-creating flows	-0.2	2.6	7.8			12.7	9.7	7.7	5.8	3.4	2.7		0.1	0.6	
Non-interest current account deficit	**24.3**	**25.9**	**28.1**	**21.3**	**11.6**	**29.9**	**31.3**	**28.9**	**25.8**	**23.6**	**22.7**		**25.1**	**20.6**	24.0
Deficit in balance of goods and services	77.3	66.2	71.9			87.3	86.7	70.4	55.7	52.2	48.2		40.5	29.8	
Exports	48.3	47.5	42.6			38.1	32.5	27.5	24.3	23.7	23.2		25.9	22.9	
Imports	125.6	113.8	114.5			125.4	119.2	97.9	80.1	75.8	71.3		66.4	52.7	
Net current transfers (negative = inflow)	-66.5	-57.4	-61.7	-99.6	38.2	-73.1	-68.1	-53.3	-42.2	-39.5	-36.5		-25.6	-15.0	-22.4
of which: official	-26.7	-25.1	-32.2			-37.6	-32.3	-28.7	-26.1	-24.1	-22.0		-14.2	-8.1	
Other current account flows (negative = net inflow)	13.5	17.1	17.9			15.7	12.7	11.8	12.3	10.9	11.0		10.2	5.7	
Net FDI (negative = inflow)	**-22.8**	**-22.3**	**-19.6**	**-13.7**	**17.2**	**-17.7**	**-21.3**	**-21.1**	**-19.0**	**-18.5**	**-18.3**		**-22.7**	**-18.8**	-21.7
Endogenous debt dynamics 2/	**-1.6**	**-0.9**	**-0.7**			**0.5**	**-0.4**	**-0.2**	**-1.0**	**-1.6**	**-1.7**		**-2.3**	**-1.1**	
Contribution from nominal interest rate	0.0	0.1	0.1			0.1	0.2	0.2	0.2	0.3	0.3		0.3	0.2	
Contribution from real GDP growth	-0.8	-0.7	-0.6			0.4	-0.5	-0.4	-1.2	-1.9	-2.0		-2.6	-1.4	
Contribution from price and exchange rate changes	-0.8	-0.4	-0.2			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-2.1**	**-1.2**	**-5.1**			**-2.9**	**-3.7**	**-2.0**	**-2.8**	**-3.0**	**-3.3**		**-0.4**	**-1.1**	
of which: exceptional financing	0.0	0.0	0.0			-0.9	-0.9	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	8.1			14.5	17.7	21.2	23.2	23.4	22.9		20.2	17.6	
In percent of exports	...	...	19.0			38.1	54.4	77.2	95.1	98.9	98.8		77.9	76.9	
PV of PPG external debt	**...**	**...**	**8.1**			**14.5**	**17.7**	**21.2**	**23.2**	**23.4**	**22.9**		**20.2**	**17.6**	
In percent of exports	**...**	**...**	**19.0**			**38.1**	**54.4**	**77.2**	**95.1**	**98.9**	**98.8**		**77.9**	**76.9**	
In percent of government revenues	**...**	**...**	**34.4**			**64.6**	**80.9**	**92.9**	**100.2**	**97.3**	**92.6**		**83.8**	**66.0**	
Debt service-to-exports ratio (in percent)	**0.1**	**0.8**	**0.4**			**0.6**	**0.9**	**1.2**	**1.8**	**4.0**	**5.9**		**4.2**	**4.4**	
PPG debt service-to-exports ratio (in percent)	**0.1**	**0.8**	**0.4**			**0.6**	**0.9**	**1.2**	**1.8**	**3.9**	**5.9**		**4.2**	**4.4**	
PPG debt service-to-revenue ratio (in percent)	**0.2**	**1.4**	**0.8**			**1.1**	**1.3**	**1.5**	**1.9**	**3.9**	**5.5**		**4.5**	**3.8**	
Total gross financing need (Millions of U.S. dollars)	24.2	73.5	175.9			248.3	214.0	179.1	168.3	153.3	165.8		153.3	286.9	
Non-interest current account deficit that stabilizes debt ratio	26.5	24.4	25.4			20.1	25.4	23.3	22.8	23.2	23.3		25.4	21.1	
Key macroeconomic assumptions															
Real GDP growth (in percent)	7.9	8.5	5.7	7.1	1.8	-2.8	2.4	1.4	3.7	5.5	5.9	2.7	7.9	4.8	6.8
GDP deflator in US dollar terms (change in percent)	7.6	4.0	2.3	6.3	2.7	2.2	1.6	3.9	2.7	4.2	4.9	3.2	1.1	3.7	2.2
Effective interest rate (percent) 5/	0.1	1.6	0.7	0.4	0.6	0.9	0.7	0.7	0.7	0.8	0.8	0.8	1.0	0.8	0.9
Growth of exports of G&S (US dollar terms, in percent)	26.1	11.1	-3.1	15.0	16.8	-11.3	-11.1	-10.9	-5.6	6.8	8.8	-3.9	11.2	4.7	9.1
Growth of imports of G&S (US dollar terms, in percent)	9.3	2.2	8.8	15.7	24.0	8.8	-1.0	-13.5	-12.8	4.0	4.5	-1.7	5.4	5.0	7.0
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	47.1	54.5	45.9	47.9	50.6	50.7	49.4	51.1	54.0	53.0
Government revenues (excluding grants, in percent of GDP)	26.2	27.5	23.5			22.5	21.8	22.8	23.1	24.0	24.7		24.1	26.6	25.5
Aid flows (in Millions of US dollars) 7/	28.3	45.7	77.6			199.1	198.6	196.9	104.3	76.0	81.7		88.9	209.7	
of which: Grants	28.3	45.7	77.6			199.1	198.6	196.9	104.3	76.0	81.7		88.9	209.7	
of which: Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			15.5	12.9	12.3	7.4	5.3	5.1		3.7	3.5	3.8
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			71.5	82.5	76.0	70.0	69.7	70.4		69.2	73.3	71.5
Memorandum items:															
Nominal GDP (Millions of US dollars)	1643.1	1853.7	2004.5			1991.1	2072.8	2183.0	2325.7	2555.9	2838.6		4446.1	10485.1	
Nominal dollar GDP growth	16.0	12.8	8.1			-0.7	4.1	5.3	6.5	9.9	11.1	6.0	9.1	8.7	9.1
PV of PPG external debt (in Millions of US dollars)			157.3			283.8	348.7	443.8	529.0	590.9	642.2		885.3	1815.6	
(PVt-PVt-1)/GDPt-1 (in percent)						**6.3**	3.3	4.6	3.9	2.7	2.0	3.8	1.6	1.2	1.4
Gross workers' remittances (Millions of US dollars)	...	...	...			...	...	...	...	...	...		...	...	
PV of PPG external debt (in percent of GDP + remittances)	...	...	8.1			14.5	17.7	21.2	23.2	23.4	22.9		20.2	17.6	
PV of PPG external debt (in percent of exports + remittances)	...	...	19.0			38.1	54.4	77.2	95.1	98.9	98.8		77.9	76.9	
Debt service of PPG external debt (in percent of exports + remittance	...	...	0.4			0.6	0.9	1.2	1.8	3.9	5.9		4.2	4.4	

Sources: Country authorities; and staff estimates and projections.

1/ Includes only public. Fiscal year (July 1st to June 30th).

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table A2. Liberia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2012–35

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2012	2013	2014	Average 6/	Standard Deviation 6/	2015	2016	2017	2018	2019	2020	2015-20 Average	2025	2035	2021-35 Average
Public sector debt 1/	11.0	12.4	16.1			24.3	29.9	40.3	43.7	40.9	40.0		36.0	31.3	
of which: foreign-currency denominated	11.0	12.4	16.1			24.3	29.9	37.1	40.9	40.9	40.0		35.9	31.3	
Change in public sector debt	-2.6	1.4	3.8			8.1	5.6	10.4	3.5	-2.8	-0.9		-0.5	-0.6	
Identified debt-creating flows	1.3	0.6	1.0			8.3	4.0	5.8	2.5	0.2	-0.7		-1.3	-1.7	
Primary deficit	3.2	1.4	1.8	0.0	2.1	8.2	4.0	7.1	5.2	3.8	2.9	5.2	1.4	0.6	1.2
Revenue and grants	27.9	29.9	27.4			32.5	31.4	31.8	27.6	27.0	27.6		26.1	28.6	
of which: grants	1.7	2.5	3.9			10.0	9.6	9.0	4.5	3.0	2.9		2.0	2.0	
Primary (noninterest) expenditure	31.2	31.4	29.2			40.7	35.4	38.9	32.8	30.9	30.5		27.5	29.2	
Automatic debt dynamics	-1.9	-0.8	-0.8			0.1	0.0	-1.2	-2.7	-3.6	-3.6		-2.7	-2.3	
Contribution from interest rate/growth differential	-1.2	-0.9	-0.7			0.5	-0.6	-0.4	-1.7	-2.7	-2.6		-3.0	-1.8	
of which: contribution from average real interest rate	-0.2	0.0	0.0			0.0	-0.1	0.0	-0.2	-0.4	-0.3		-0.3	-0.3	
of which: contribution from real GDP growth	-1.0	-0.9	-0.7			0.5	-0.6	-0.4	-1.5	-2.3	-2.3		-2.7	-1.4	
Contribution from real exchange rate depreciation	-0.7	0.1	-0.1			-0.3	0.7	-0.8	-1.0	-0.9	-1.0		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes 2/	-3.9	0.8	2.8			-0.2	1.6	4.5	0.9	-3.0	-0.3		0.8	1.2	
Other Sustainability Indicators															
PV of public sector debt	...	...	10.6			15.4	18.2	26.4	28.8	25.8	25.0		21.1	17.7	
of which: foreign-currency denominated	...	...	10.6			15.4	18.2	23.3	26.0	25.8	24.9		21.1	17.7	
of which: external	...	...	8.1			14.5	17.7	21.2	23.2	23.4	22.9		20.2	17.6	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 3/	3.4	2.3	2.1			9.2	4.7	8.2	6.3	5.2	4.5		2.6	1.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	38.8			47.3	57.9	83.0	104.5	95.6	90.4		80.9	61.9	
PV of public sector debt-to-revenue ratio (in percent)	...	...	45.2			68.4	83.3	115.8	124.8	107.5	100.9		87.6	66.6	
of which: external 4/	...	...	34.4			64.6	80.9	92.9	100.2	97.3	92.6		83.8	66.0	
Debt service-to-revenue and grants ratio (in percent) 5/	0.4	3.0	1.1			3.0	2.2	3.7	4.0	5.1	5.9		4.7	3.7	
Debt service-to-revenue ratio (in percent) 5/	0.5	3.2	1.2			4.4	3.1	5.2	4.7	5.7	6.6		5.1	4.0	
Primary deficit that stabilizes the debt-to-GDP ratio	5.8	0.1	-2.0			0.1	-1.6	-3.3	1.7	6.7	3.8		1.9	1.1	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.9	8.5	5.7	7.1	1.8	-2.8	2.4	1.4	3.7	5.5	5.9	2.7	7.9	4.8	6.8
Average nominal interest rate on forex debt (in percent)	0.2	1.4	1.3	0.6	0.6	1.3	1.0	1.2	1.4	1.4	1.3	1.3	1.1	0.9	1.1
Average real interest rate on domestic debt (in percent)	-5.7	-6.6	...	-5.8	4.4	...	...	714.5	1.8	-2.1	1.7	...	2.3	-0.7	...
Real exchange rate depreciation (in percent, + indicates depreciation)	-5.6	0.7	-0.8	-3.6	4.1	-2.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	8.7	8.0	9.2	9.5	3.5	7.3	9.0	14.1	9.0	7.2	7.3	9.0	4.4	7.0	5.7
Growth of real primary spending (deflated by GDP deflator, in percent)	24.5	9.1	-1.7	3.4	8.0	35.4	-10.8	11.3	-12.5	-0.8	4.7	4.6	8.2	3.8	6.5
Grant element of new external borrowing (in percent)	...	...	...	...	...	47.1	54.5	45.9	47.9	50.6	50.7	49.4	51.1	54.0	...

Sources: Country authorities; and staff estimates and projections.

1/ Central government. Fiscal year (July 1st to June 30th) base.

2/ The government issued two-year T-bonds of L$6 billion in FY2017 for liquidity purpose. Collected money was deposited in the CBL account. These T-bonds are included in public debt, and transactions are captured in "Residual, including asset changes."

3/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

4/ Revenues excluding grants.

5/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table A3. Liberia: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015–35

(Percent)

	Projections							
	2015	2016	2017	2018	2019	2020	**2025**	2035
PV of debt-to GDP ratio								
Baseline	15	18	21	23	23	23	**20**	18
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	15	14	15	15	15	15	**17**	21
A2. New public sector loans on less favorable terms in 2015-2035 2/	15	18	24	27	29	29	**30**	30
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	15	16	19	21	22	21	**19**	16
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	15	16	18	21	21	21	**18**	17
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	15	17	20	23	23	23	**20**	17
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	15	22	23	25	26	25	**21**	18
B5. Combination of B1-B4 using one-half standard deviation shocks	15	11	-2	1	2	3	**5**	11
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	15	24	29	32	33	32	**28**	25
PV of debt-to-exports ratio								
Baseline	38	54	77	95	99	99	**78**	77
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	38	42	53	61	63	64	**67**	92
A2. New public sector loans on less favorable terms in 2015-2035 2/	38	57	86	112	121	125	**115**	129
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	38	52	74	93	98	98	**77**	76
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	38	47	59	76	80	81	**64**	66
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	38	52	74	93	98	98	**77**	76
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	38	67	84	104	108	108	**83**	78
B5. Combination of B1-B4 using one-half standard deviation shocks	38	31	-6	3	8	10	**17**	42
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	38	52	74	93	98	98	**77**	76
PV of debt-to-revenue ratio								
Baseline	65	81	93	100	97	93	**84**	66
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	65	63	64	64	62	60	**72**	79
A2. New public sector loans on less favorable terms in 2015-2035 2/	65	84	103	118	119	117	**123**	111
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	65	75	83	92	90	85	**77**	61
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	65	73	79	89	87	84	**77**	63
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	65	76	89	98	96	91	**82**	65
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	65	100	101	110	106	101	**89**	67
B5. Combination of B1-B4 using one-half standard deviation shocks	65	49	-8	3	9	11	**21**	42
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	65	109	126	140	136	130	**117**	92

Table A3. Liberia: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2015–35 (Concluded)

(Percent)

Debt service-to-exports ratio								
Baseline	1	1	1	2	4	6	**4**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	1	1	1	1	3	4	**3**	3
A2. New public sector loans on less favorable terms in 2015-2035 2/	1	1	1	3	5	7	**4**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	1	1	1	2	4	6	**4**	4
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	1	1	1	2	3	5	**3**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	1	1	1	2	4	6	**4**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	1	1	2	2	4	6	**5**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	1	1	1	0	2	4	**0**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	1	1	1	2	4	6	**4**	4
Debt service-to-revenue ratio								
Baseline	1	1	1	2	4	6	**5**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2015-2035 1/	1	1	1	1	3	4	**3**	3
A2. New public sector loans on less favorable terms in 2015-2035 2/	1	1	1	3	5	7	**4**	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2016-2017	1	1	1	2	4	5	**4**	3
B2. Export value growth at historical average minus one standard deviation in 2016-2017 3/	1	1	1	2	4	5	**4**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2016-2017	1	1	1	2	4	5	**5**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2016-2017 4/	1	1	2	2	4	6	**5**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	1	1	1	0	2	4	**0**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2016 5/	1	2	2	3	6	8	**6**	5
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	51	51	51	51	51	51	**51**	51

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table A4. Liberia: Sensitivity Analysis for Key Indicators of Public Debt, 2015–35

		Projections						
	2015	2016	2017	2018	2019	2020	2025	2035
PV of Debt-to-GDP Ratio								
Baseline	15	18	26	29	26	25	21	18
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	15	16	19	19	15	13	9	7
A2. Primary balance is unchanged from 2015	15	20	29	33	31	33	41	57
A3. Permanently lower GDP growth 1/	15	18	27	29	27	26	24	27
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	15	17	24	25	22	21	14	7
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	15	17	23	26	23	22	19	17
B3. Combination of B1-B2 using one half standard deviation shocks	15	16	21	22	19	17	11	4
B4. One-time 30 percent real depreciation in 2016	15	24	31	32	28	27	21	16
B5. 10 percent of GDP increase in other debt-creating flows in 2016	15	23	31	33	30	29	24	19
PV of Debt-to-Revenue Ratio 2/								
Baseline	47	58	83	105	96	90	81	62
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	47	51	63	70	56	49	36	24
A2. Primary balance is unchanged from 2015	47	64	91	118	116	118	158	200
A3. Permanently lower GDP growth 1/	47	58	84	106	99	94	92	93
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	47	56	76	93	82	75	55	25
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	47	55	73	94	85	81	73	58
B3. Combination of B1-B2 using one half standard deviation shocks	47	52	66	81	70	63	42	13
B4. One-time 30 percent real depreciation in 2016	47	78	98	117	105	96	80	57
B5. 10 percent of GDP increase in other debt-creating flows in 2016	47	72	97	120	110	104	92	67
Debt Service-to-Revenue Ratio 2/								
Baseline	3	2	4	4	5	6	5	4
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	3	2	3	3	4	5	3	2
A2. Primary balance is unchanged from 2015	3	2	4	4	5	6	6	8
A3. Permanently lower GDP growth 1/	3	2	4	4	5	6	5	5
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2016-2017	3	2	4	4	5	5	4	2
B2. Primary balance is at historical average minus one standard deviations in 2016-2017	3	2	4	4	5	6	4	3
B3. Combination of B1-B2 using one half standard deviation shocks	3	2	3	3	4	5	4	2
B4. One-time 30 percent real depreciation in 2016	3	2	4	5	7	9	7	6
B5. 10 percent of GDP increase in other debt-creating flows in 2016	3	2	4	5	5	6	6	4

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.